

```

maxFollow <- 84
cens.type <- "weibull"

# m1 -censoring adjustment
muC.adj <- log(1.5)
k.z3 <- 1
k.treat <- 0.9

z1_frac <- 0.25 # Default model index 'm1' (The 1st quartile of z1=er)
pH_super <- 0.125 # non-NULL re-defines z1_frac

if (is.null(pH_super)) {
  # pH_check<-with(gbsg,mean(pgr<=quantile(pgr,c(z3_frac),1,0) &
  # er<=quantile(er,z1_frac)))
  pH_check <- with(gbsg, mean(meno == 0 & er <= quantile(er, z1_frac)))
  cat("Underlying pH_super", c(pH_check), "\n")
}
# pH_super specified If pH_super then override z1_frac and find z1_frac to
# yield pH_super

if (!is.null(pH_super)) {
  # Approximate Z1 quantile to yield pH proportion
  z1_q <- uniroot(propH.obj4, c(0, 1), tol = 1e-04, pH.target = pH_super)$root
  # pH_check<-with(gbsg,mean(pgr<=quantile(pgr,c(z3_frac),1,0) &
  # er<=quantile(er,z1_q)))
  pH_check <- with(gbsg, mean(meno == 0 & er <= quantile(er, z1_q)))
  cat("pH", c(pH_check), "\n")
  rel_error <- (pH_super - pH_check)/pH_super
  if (abs(rel_error) >= 0.1)
    stop("pH_super approximation relative error exceeds 10%")
  z1_frac <- z1_q
  cat("Underlying pH_super", c(pH_check), "\n")
}

## pH 0.122449
## Underlying pH_super 0.122449

# Bootstrap on log(hr) scale converted to HR (est.loghr=TRUE & est.scale='hr')
t.start.all <- proc.time()[3]

```

```

##### Forest search criteria
hr.threshold <- 1.25 # Initital candidates
hr.consistency <- 1 # Candidates for many splits
pconsistency.threshold <- 0.9
stop.threshold <- 0.95
maxk <- 2
nmin.fs <- 60
pstop_futile <- 0.5
# Limit timing for forestsearch
max.minutes <- 3
m1.threshold <- Inf # Turning this off (Default)
# pconsistency.threshold<-0.70 # Minimum threshold (will choose max among
# subgroups satisfying)
fs.splits <- 400 # How many times to split for consistency
# vi is % factor is selected in cross-validation --> higher more important

```

```
vi.grf.min <- 0.2
# Null, turns off grf screening
d.min <- 10 # Min number of events for both arms (d0.min=d1.min=d.min)
# default=5 Virtual twins analysis Counter-factual difference (C-E) >=
# vt.threshold Large values in favor of C (control)
vt.threshold <- 0.225 # For VT delta
treat.threshold <- 0

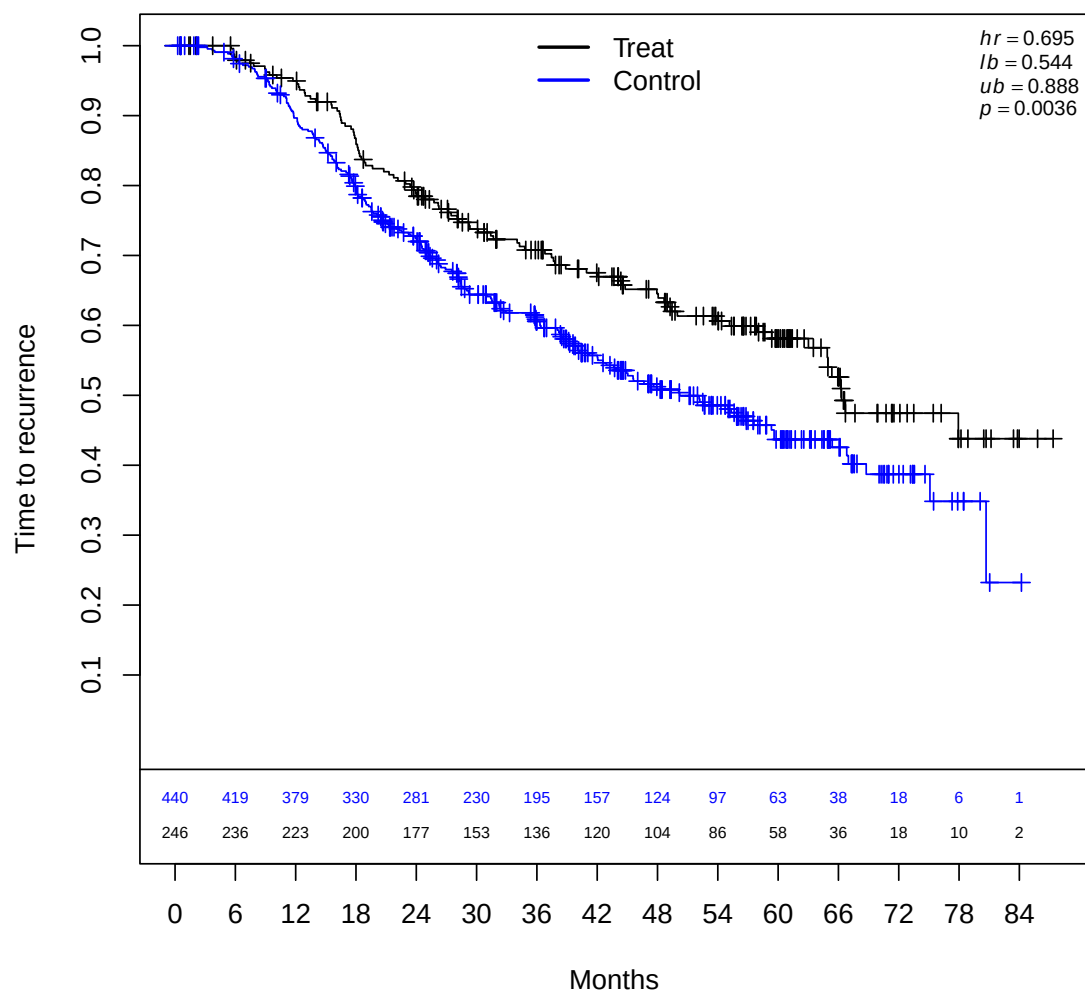
maxdepth <- 2
n.min <- 60
ntree <- 1000
# GRF criteria
dmin.grf <- 12 # For GRF delta
# Note: For CRT this represents dmin.grf/2 RMS for control (-dmin.grf/2 for
# treatment)
frac.tau <- 0.6

outcome.name <- c("y.sim")
event.name <- c("event.sim")
id.name <- c("id")
treat.name <- c("treat")

cox.formula.sim <- as.formula(paste("Surv(y.sim,event.sim)~treat"))
cox.formula.adj.sim <- as.formula(paste("Surv(y.sim,event.sim)~treat+v1+v2+v3+v4+v5"))

mod.harm <- "alt"
hrH.target <- 2

# out.loc = NULL turns off file creation
N <- 700
this.dgm <- get.dgm4(mod.harm = mod.harm, N = N, k.treat = k.treat, hrH.target = hrH.target,
  cens.type = cens.type, out.loc = NULL, details = TRUE, parms_torand = FALSE)
```



```
## Super-population empirical harm and non-harm hazard ratios= 2.000007 0.6466405
## Causal HR (empirical ITT)= 0.7057463
```

```
n_add_noise <- 3
sim <- 1

dgm <- this.dgm$dgm

x <- sim_aftm4_gbsg(dgm = dgm, n = N, maxFollow = maxFollow, muC.adj = muC.adj, simid = sim)

if (n_add_noise == 0) {
  confounders.name <- c("z1", "z2", "z3", "z4", "z5", "size", "grade")
}

if (n_add_noise == 5) {
  set.seed(8316951 + 1000 * sim)
  # Add 5 noise
}
```

```

x$noise1 <- rnorm(N)
x$noise2 <- rnorm(N)
x$noise3 <- rnorm(N)
x$noise4 <- rnorm(N)
x$noise5 <- rnorm(N)
confounders.name <- c("z1", "z2", "z3", "z4", "z5", "size", "grade", "noise1",
  "noise2", "noise3", "noise4", "noise5")
}

if (n_add_noise == 3) {
  set.seed(8316951 + 1000 * sim)
  # Add 3 noise
  x$noise1 <- rnorm(N)
  x$noise2 <- rnorm(N)
  x$noise3 <- rnorm(N)
  confounders.name <- c("z1", "z2", "z3", "z4", "z5", "size", "grade", "noise1",
    "noise2", "noise3")
}

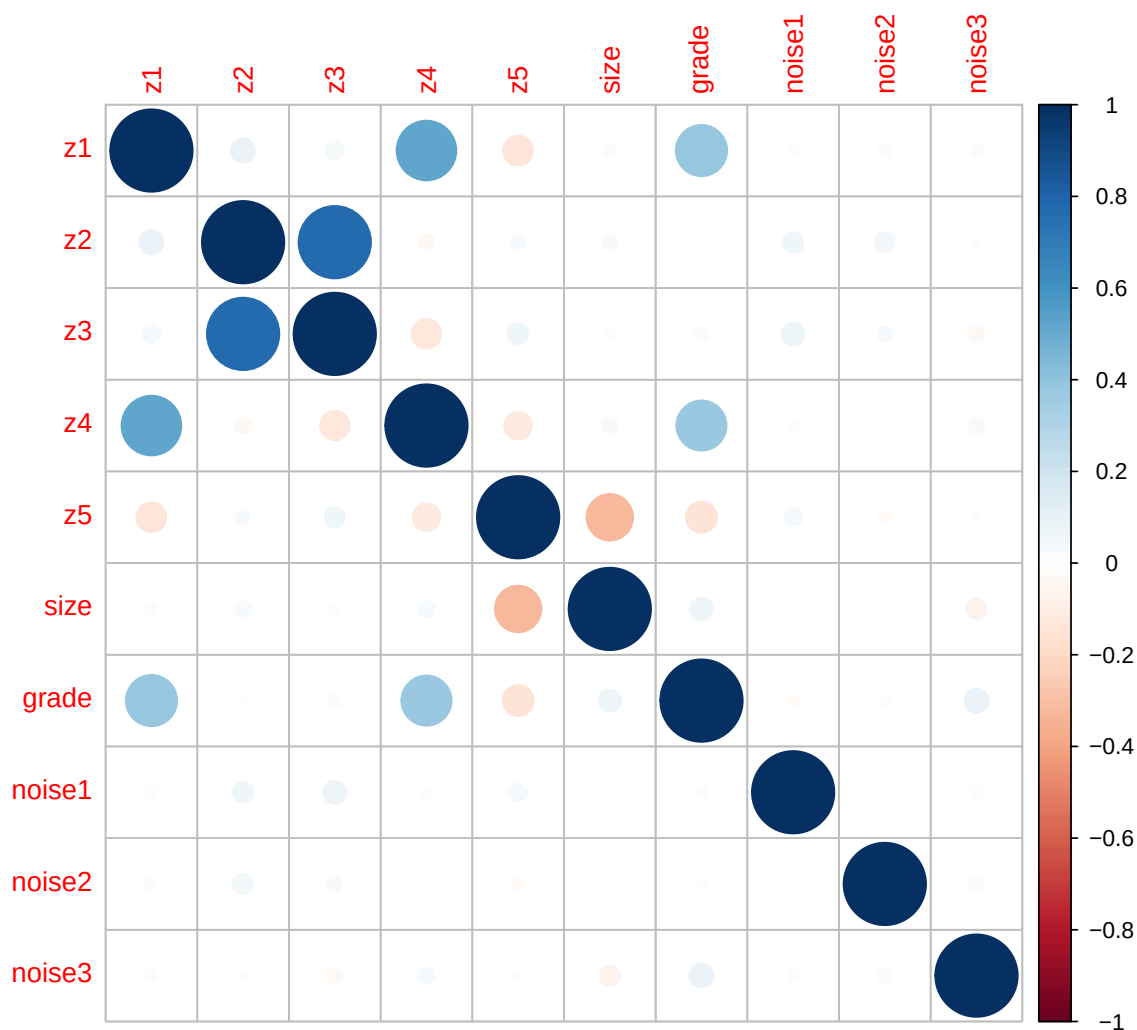
# More challenging which_replace <- which(confounders.name=='z1')
# confounders.name[which_replace] <- 'er'

cox.formula.check <- as.formula(paste("Surv(y.sim,event.sim)~treat+v1+v2+v3+v4+v5+size+grade+noise1+
coxph(cox.formula.check, data = x)

## Call:
## coxph(formula = cox.formula.check, data = x)
##
##              coef exp(coef) se(coef)      z      p
## treat  -0.190994  0.826138  0.106954 -1.786  0.07414
## v11      0.646240  1.908352  0.139462  4.634 3.59e-06
## v21     -0.644920  0.524705  0.199045 -3.240  0.00119
## v31      0.910511  2.485592  0.202101  4.505 6.63e-06
## v41      0.601285  1.824462  0.134425  4.473 7.71e-06
## v51     -0.823439  0.438920  0.112756 -7.303 2.82e-13
## size      0.003801  1.003809  0.003565  1.066  0.28623
## grade   -0.042503  0.958388  0.100926 -0.421  0.67366
## noise1  -0.046821  0.954258  0.054690 -0.856  0.39193
## noise2   0.130029  1.138861  0.054183  2.400  0.01640
## noise3   0.027800  1.028190  0.052403  0.531  0.59576
##
## Likelihood ratio test=188.3 on 11 df, p=< 2.2e-16
## n= 700, number of events= 365

Zm <- cor(x[, c(confounders.name)])
corrplot(Zm)

```



```
use_lasso <- TRUE
use_grf <- TRUE
use_grf_only <- TRUE
```

```
fs.est <- forestsearch(df.analysis = x, Allconfounders.name = confounders.name, details = TRUE,
  use_lasso = use_lasso, use_grf = use_grf, use_grf_only = use_grf_only, conf_force = NULL,
  outcome.name = outcome.name, treat.name = treat.name, event.name = event.name,
  id.name = id.name, n.min = nmin.fs, hr.threshold = hr.threshold, hr.consistency = hr.consistency,
  fs.splits = fs.splits, d0.min = d.min, d1.min = d.min, pstop_futile = pstop_futile,
  pconsistency.threshold = pconsistency.threshold, stop.threshold = stop.threshold,
  max.minutes = max.minutes, maxk = maxk, by.risk = 12, plot.sg = TRUE, vi.grf.min = vi.grf.min)
```

```
## tau, maxdepth= 77.92977 2
```

```
##   leaf.node control.mean control.size control.se treated.mean treated.size
## 1         2   -10.725803   325.000000   2.824359    10.725803   325.000000
## 2         3    2.742131   375.000000   2.528028    -2.742131   375.000000
## 3         4   -17.438997   150.000000   4.105683    17.438997   150.000000
## 4         5    7.149639   152.000000   3.980382    -7.149639   152.000000
```

```

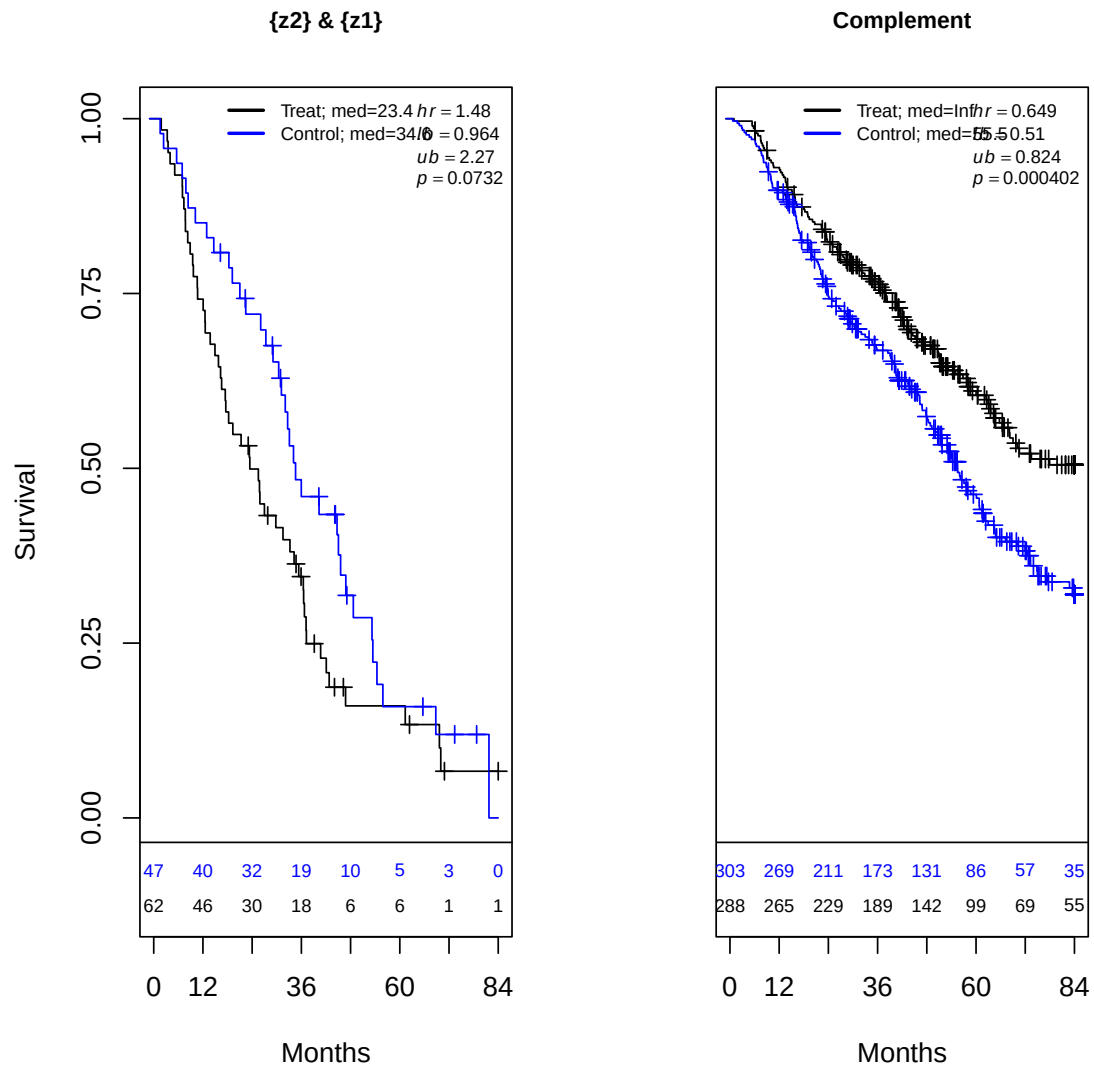
## 5      6      7.949972    146.000000    4.002286    -7.949972    146.000000
## 6      7     -8.290389    252.000000    3.123463     8.290389    252.000000
##   treated.se      diff Nsg depth
## 1   2.824359 -21.451606 325      1
## 2   2.528028   5.484263 375      1
## 3   4.105683 -34.877994 150      2
## 4   3.980382  14.299278 152      2
## 5   4.002286  15.899943 146      2
## 6   3.123463 -16.580779 252      2
##   leaf.node control.mean control.size control.se treated.mean treated.size
## 5      6      7.949972    146.000000    4.002286    -7.949972    146.000000
##   treated.se      diff Nsg depth
## 5   4.002286 15.89994 146      2
## [1] "noise2 <= -0.12" "z2 <= 0"          "noise1 <= -0.42"
## [1] "noise1 <= -0.419923509251078"
## # of continuous/categorical characteristics 4 6
## Continuous characteristics: size noise1 noise2 noise3
## Categorical characteristics: z1 z2 z3 z4 z5 grade
## Initial GRF cuts included noise2 <= -0.12 z2 <= 0 noise1 <= -0.42
## # of candidate subgroup factors= 8
## [1] "noise2 <= -0.12" "noise1 <= -0.42" "z1"          "z2"
## [5] "z3"          "z4"          "z5"          "grade"
## Confounders per grf screening q4 q8 q5 q1 q3 q2 q7 q6
##   FSconfounders.name      vi.cs vi_ratio
## 4      q4 0.27260018 2.1808015
## 8      q8 0.14615634 1.1692507
## 5      q5 0.13149717 1.0519773
## 1      q1 0.09785155 0.7828124
## 3      q3 0.09315877 0.7452701
## 2      q2 0.09121612 0.7297290
## 7      q7 0.08435687 0.6748550
## 6      q6 0.08316299 0.6653039
## Number of unique levels (L) and possible subgroups= 17 131071
## # of subgroups based on # variables > k.max and excluded (per million) 0.130918
## k.max= 2
## Events criteria for control,exp= 10 10
## # of subgroups with events less than criteria: control, experimental 21 18
## # of subgroups with sample size less than criteria 24
## # of subgroups meeting all criteria = 128
## # of subgroups fitted (Cox model estimable) = 128
## *Subgroup Searching Minutes=* 0.0158
## Number of subgroups meeting HR threshold 11
## # of candidate subgroups (meeting HR criteria) = 11
## Subgroups (1st 10) meeting overall screening thresholds (HR, m1) sorted by HRs= Inf
##   K   n   E d1   m1   m0   HR L(HR) U(HR) q4.0 q4.1 q8.1 q8.2 q8.3 q5.0 q5.1
## 1: 2   62  28 15 56.82   NA 2.02 0.95 4.28   0   0   1   0   0   0   0
## 2: 2   77  30 15 64.35   NA 1.61 0.78 3.30   0   0   1   0   0   0   0
## 3: 2   63  22 12 68.24   NA 1.57 0.67 3.65   0   0   1   0   0   0   0
## 4: 2  131  73 44 46.30 60.76 1.52 0.95 2.43   0   1   0   0   0   0   0
## 5: 2   85  74 45 19.29 32.07 1.51 0.94 2.41   0   0   0   0   0   0   1
## 6: 1   89  37 19 63.58   NA 1.50 0.78 2.87   0   0   1   0   0   0   0
## 7: 2  146  84 46 40.70 50.63 1.49 0.97 2.30   0   0   0   0   0   0   0
## 8: 2  109  88 52 23.44 34.58 1.48 0.96 2.27   0   1   0   0   0   0   0
## 9: 2   98  58 34 42.07 54.22 1.38 0.82 2.34   0   0   0   0   0   0   1
## 10: 2   61  51 26 26.99 39.47 1.37 0.78 2.41   0   0   0   0   0   0   0

```

```

##      q1.0 q1.1 q3.0 q3.1 q2.0 q2.1 q7.0 q7.1 q6.0 q6.1
## 1:      0      0      0      0      1      0      0      0      0      0
## 2:      0      0      0      0      0      0      0      0      1      0
## 3:      0      0      0      0      0      0      0      1      0      0
## 4:      0      0      0      0      0      1      0      0      0      0
## 5:      0      0      0      1      0      0      0      0      0      0
## 6:      0      0      0      0      0      0      0      0      0      0
## 7:      1      0      0      0      0      1      0      0      0      0
## 8:      0      0      0      1      0      0      0      0      0      0
## 9:      0      0      0      0      0      1      0      0      0      0
## 10:     0      0      0      1      0      1      0      0      0      0
## Consistency 0.935
## # of splits= 400
## Model, % Consistency Met= {grade} !{noise1 <= -0.42} 0.935
## Consistency 0.7875
## Consistency 0.66
## Consistency 0.8975
## Consistency 0.8975
## Consistency 0.785
## Consistency 0.9275
## # of splits= 400
## Model, % Consistency Met= !{noise2 <= -0.12} {noise1 <= -0.42} 0.9275
## Consistency 0.955
## # of splits= 400
## Model, % Consistency Met= {z2} {z1} 0.955
## Number of subgroups meeting consistency criteria=
##      p.consistency Nsg group.id m.index K          M.1          M.2
## 1:      0.9550 109          3          8 2          {z2}          {z1}
## 2:      0.9350 62          4          1 2          {grade} !{noise1 <= -0.42}
## 3:      0.9275 146          6          7 2 !{noise2 <= -0.12} {noise1 <= -0.42}

```



```
## [1] "{z2}" "{z1}"
## % consistency criteria met= 0.955
## SG focus= hr
## Subgroup Consistency Minutes= 0.1766167
## Subgroup found
## Minutes overall= 0.2098667

grf.est <- grf.subg.harm.survival(data = x, confounders.name = confounders.name,
  outcome.name = outcome.name, event.name = event.name, id.name = id.name, treat.name = treat.name,
  n.min = n.min, dmin.grf = dmin.grf, frac.tau = 1, details = TRUE)

## tau, maxdepth= 77.92977 2
##   leaf.node control.mean control.size control.se treated.mean treated.size
## 1         2  -10.535829   325.000000   2.834699   10.535829   325.000000
## 2         3    2.727493   375.000000   2.515591   -2.727493   375.000000
## 3         4  -17.261313   150.000000   4.120255   17.261313   150.000000
## 4         5    7.069757   152.000000   3.966572   -7.069757   152.000000
## 5         6    8.102285   146.000000   4.006352   -8.102285   146.000000
## 6         7   -8.212993   252.000000   3.113917    8.212993   252.000000
```

```
##   treated.se      diff Nsg depth
## 1   2.834699 -21.071658 325     1
## 2   2.515591   5.454986 375     1
## 3   4.120255 -34.522626 150     2
## 4   3.966572  14.139513 152     2
## 5   4.006352  16.204571 146     2
## 6   3.113917 -16.425985 252     2
##   leaf.node control.mean control.size control.se treated.mean treated.size
## 5         6      8.102285   146.000000   4.006352    -8.102285   146.000000
##   treated.se      diff Nsg depth
## 5   4.006352 16.20457 146     2
## [1] "noise2 <= -0.12" "z2 <= 0"          "noise1 <= -0.42"
## [1] "noise1 <= -0.419923509251078"
```

```
library(doParallel)
registerDoParallel(parallel::detectCores(logical = FALSE))

cox.formula.boot <- cox.formula.sim
max.minutes <- 7
# Suggest running 50, first ... to get timing estimate

NB <- 20

df_boot_analysis <- fs.est$df.est

fitH <- get_Cox_sg(df_sg = subset(df_boot_analysis, treat.recommend == 0), cox.formula = cox.formula)
H_obs <- fitH$est_obs # log(hr) scale
seH_obs <- fitH$se_obs
# Hc observed estimates
fitHc <- get_Cox_sg(df_sg = subset(df_boot_analysis, treat.recommend == 1), cox.formula = cox.formula)
Hc_obs <- fitHc$est_obs
seHc_obs <- fitHc$se_obs
rm("fitH", "fitHc")

Ystar_mat <- bootYstar({
  ystar <- get_Ystar(boot)
}, boots = NB, seed = 8316951, counter = "boot", export = fun_arg_list_boot)
# Check dimension
if (dim(Ystar_mat)[1] != NB | dim(Ystar_mat)[2] != nrow(df_boot_analysis)) stop("Dimension of Ystar_mat")

tB.start <- proc.time()[3]
# Bootstraps
resB <- bootPar({
  ans <- fsboot_forparallel(boot)
}, boots = NB, seed = 8316951, counter = "boot", export = fun_arg_list_boot)
tB.now <- proc.time()[3]
tB.min <- (tB.now - tB.start)/60

doParallel::stopImplicitCluster()

cat("Minutes for Boots", c(NB, tB.min), "\n")

## Minutes for Boots 20 0.7836333

cat("Projection per 100", c(tB.min * (100/NB)), "\n")
```

```

## Projection per 100 3.918167

cat("Propn bootstrap subgroups found =", c(sum(!is.na(resB$H_biasadj_1))/NB), "\n")

## Propn bootstrap subgroups found = 1

# How many timmed out
cat("Number timmed out=", c(sum(is.na(resB$H_biasadj_1) & resB$tmins_search > max.minutes)),
    "\n")

## Number timmed out= 0

H_estimates <- get_dfRes(Hobs = H_obs, seHobs = seH_obs, H1_adj = resB$H_biasadj_1,
    ystar = Ystar_mat, cov_method = "standard", cov_trim = 0.05)

Hc_estimates <- get_dfRes(Hobs = Hc_obs, seHobs = seHc_obs, H1_adj = resB$Hc_biasadj_1,
    ystar = Ystar_mat, cov_method = "standard", cov_trim = 0.05)

print(H_estimates)

##           H0          sdH0  H0_lower H0_upper          H1          sdH1  H1_lower H1_upper
## 1: 1.478681 0.3227903 0.9639631 2.268237 1.083036 0.3461085 0.5789253 2.026112

print(Hc_estimates)

##           H0          sdH0  H0_lower H0_upper          H1          sdH1  H1_lower
## 1: 0.6485153 0.07935847 0.5102219 0.8242924 0.6494383 0.08254194 0.5062353
##           H1_upper
## 1: 0.8331505

bootit <- list(H_estimates = H_estimates, Hc_estimates = Hc_estimates)
ans3 <- na.omit(fsBoot.parallel.out(df = df_boot_analysis, FSboots = bootit, cox.formula.boot = cox.
    sim_number = sim))
tall.min <- (tB.now - t.start.all)/60
cat("Overall minutes for analysis", c(tall.min), "\n")

## Overall minutes for analysis 1.042433

# Revise path accordingly
# save(dgm,x,y,H_estimates,Hc_estimates,ans3,file='output/sim_ex1.Rdata')

```